# *Oprl1* is the dominant opioid receptor of peripheral neurons

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## Introduction

Opioid signaling in vertebrates runs through four class A G protein-coupled receptors: the mu (MOR), delta (DOR), and kappa (KOR) opioid receptors, encoded by *OPRM1*, *OPRD1*, and *OPRK1*, and the nociceptin/orphanin FQ peptide receptor (NOP), encoded by *OPRL1*. All four couple to Gi/Go and converge on a shared set of neuronal effectors: inhibition of adenylyl cyclase, activation of GIRK/Kir3 channels, and voltage-dependent suppression of Cav2.1 and Cav2.2 currents, which lowers somatic excitability and reduces transmitter release at nerve terminals (Al-Hasani and Bruchas, 2011). Selectivity arises at the orthosteric pocket. Beta-endorphin and the enkephalins act at MOR and DOR, dynorphins at KOR, and nociceptin/orphanin FQ (N/OFQ) at NOP.

NOP was cloned as an orphan receptor sharing approximately 47% amino acid identity with MOR, DOR, and KOR overall and 64% to 67% identity across the transmembrane helices (Mollereau et al., 1994; Bunzow et al., 1994). Its endogenous ligand was identified a year later (Meunier et al., 1995; Reinscheid et al., 1995). N/OFQ derives from prepronociceptin (*PNOC*) and begins with Phe-Gly-Gly-Phe in place of the Tyr-Gly-Gly-Phe motif shared by the classical opioid peptides (Mollereau et al., 1996). This substitution abolishes cross-reactivity in both directions, and naloxone and the morphinan alkaloids have negligible affinity for NOP. The antagonist-bound crystal structure accounts for the pharmacological separation: a glutamine occupies the position of the transmembrane helix 6 histidine conserved across the classical receptors, and acidic residues in extracellular loop 2 reshape the entrance to the pocket (Thompson et al., 2012). NOP therefore shares an effector repertoire with MOR, DOR, and KOR while occupying a distinct ligand space.

Peripheral opioid receptors support analgesia and visceral regulation independently of the brain. Leukocytes recruited to inflamed tissue synthesize and release POMC- and PENK-derived peptides that act on receptors at sensory terminals (Stein and Machelska, 2011). Deleting MOR or DOR selectively from primary afferents reduces opioid analgesia in inflammatory and neuropathic models (Gaveriaux-Ruff et al., 2011; Weibel et al., 2013). Drugs built on peripheral restriction are in clinical use: loperamide and the peripherally acting MOR antagonists methylnaltrexone, naloxegol, naldemedine, and alvimopan act on enteric MOR, and difelikefalin, a peripherally restricted KOR agonist, was approved in 2021 for pruritus in patients on hemodialysis (Fishbane et al., 2020). Every approved peripheral opioid drug targets MOR or KOR.

Transcriptomic surveys of sensory ganglia have sharpened this picture and exposed species differences. In human dorsal root ganglion (DRG), *OPRM1* is enriched across peptidergic nociceptor classes and *OPRD1* marks non-peptidergic and itch-associated populations, while mouse sensory neurons show no comparable cell-type partitioning (Tavares-Ferreira et al., 2022; Bhuiyan et al., 2024). *OPRK1* transcripts sit near the detection floor in human DRG, although KOR protein and KOR-mediated inhibition of voltage-gated calcium currents are demonstrable in the same tissue (Snyder et al., 2018). Evidence for peripheral NOP has accumulated separately from these atlases. NOP immunoreactivity is present in 75% to 80% of small and medium human DRG neurons and rises several-fold in suburothelial fibers from patients with detrusor overactivity and bladder pain syndrome (Anand et al., 2016). N/OFQ inhibits Cav2.2 currents in dorsal root, superior cervical, and vestibular ganglion neurons, suppresses T-type currents through a G protein-independent route (Abdulla and Smith, 1997), modulates N-type channels tonically in the absence of agonist (Beedle et al., 2004), and reduces capsaicin-evoked responses in cultured human sensory neurons at picomolar concentrations (Anand et al., 2016). NOP and N/OFQ are also present in enteric neurons, where they alter motility and secretion (Drokhlyansky et al., 2020; Toll et al., 2016). The abundance of the four receptors relative to one another across peripheral neuron types has not been measured on a common scale, and NOP has been characterized mainly as a central receptor with anti-opioid actions at supraspinal sites.

We find that *Oprl1* is the highest-expressed opioid receptor gene in neurons of the peripheral nervous system. *Oprl1* transcript abundance exceeds that of *Oprm1*, *Oprd1*, and *Oprk1* in sensory, sympathetic, parasympathetic, and enteric neuron populations, and its expression is distributed across neuronal subtypes rather than confined to nociceptors. The receptor rank order we measure in peripheral neurons inverts the rank order assumed by current peripheral opioid pharmacology. These data identify NOP as the principal opioid receptor of the mouse peripheral nervous system and as the primary candidate mediator of opioid-family signaling outside the brain.

## Results

### *Oprl1* is the highest-expressed opioid receptor gene across the peripheral nervous system

Every peripherally restricted opioid drug in clinical use acts at MOR or KOR. The four opioid receptors have rarely been measured against one another in the same peripheral neurons. We ranked *Oprl1*, *Oprm1*, *Oprd1*, and *Oprk1* within each of 19 neuronal populations from 16 mouse ganglia and plexuses, covering the sensory, sympathetic, parasympathetic, and enteric divisions. The data come from 12 published single-cell studies deposited between 2016 and 2026; we generated no new sequencing (Table S1). We took the whole-cell fraction of each deposit and applied one analysis to all of them. *Oprl1* is the highest-expressed opioid receptor gene in 18 of the 19 populations.

The ranking holds in SMART-seq full-length libraries sequenced to a median of 2.7 million reads per neuron and in droplet libraries at 1,570 UMI per cell. Separation from the other three receptors is widest in the spiral ganglion, where *Oprl1* reaches 41.94 CPM in 74.8% of neurons and *Oprm1*, *Oprd1*, and *Oprk1* are absent from all 226 cells, and in the sphenopalatine ganglion, where *Oprl1* reaches 19.50 CPM against *Oprm1* at 0.29, *Oprd1* at 0.04, and *Oprk1* at 0.01 CPM (Figures 1A and 1P). The two largest droplet datasets give the narrowest separations: *Oprl1* exceeds *Oprm1* by 1.21-fold in 26,047 nodose neurons and by 1.13-fold in 31,802 dorsal root ganglion neurons, each holding in every bootstrap replicate with a 95% confidence interval excluding 1 (Figures 1F and 1H).

The geniculate ganglion was sequenced full-length by Dvoryanchikov et al. (2017) and by droplet capture by Zhang et al. (2019), and the stellate ganglion appears in two unrelated studies. *Oprl1* ranks first in all four (Figures 1B, 1D, 1N, and 1O).

Five populations were dissected and sequenced together in one experiment (Sivori et al., 2024), which places them on a common scale. Across four sympathetic ganglia and the sphenopalatine ganglion, *Oprl1* varies 1.7-fold, from 16.33 CPM in the pelvic ganglion to 27.55 CPM in the stellate ganglion. The other three receptors vary between 6-fold and 302-fold over the same five populations. Across the full panel, *Oprm1*, *Oprk1*, and *Oprd1* each rank second in some populations, and none of them ranks second consistently. NOP is available to autonomic neurons at a similar level regardless of ganglion, and MOR, DOR, and KOR are distributed by division and cell type.

The enteric submucosal plexus before weaning ranks *Oprk1* first. At postnatal day 7, *Oprk1* reaches 122.51 CPM against 26.70 CPM for *Oprl1* in both samples at that age (Figure 1S). By postnatal day 24, *Oprk1* has fallen to 3.77 CPM at 1.1% detection, *Oprl1* stands at 35.57 CPM, and *Oprl1* ranks first in all three samples (Figure 1R). Both ages come from one deposit, one tissue, and one platform (Li et al., 2025), and the 32-fold fall in *Oprk1* is the largest change of the four receptors between them: *Oprd1* rises 7-fold, *Oprm1* falls 4-fold, and *Oprl1* moves 1.3-fold. High submucosal *Oprk1* is a property of the neonatal plexus, and the mature plexus matches the rest of the panel.

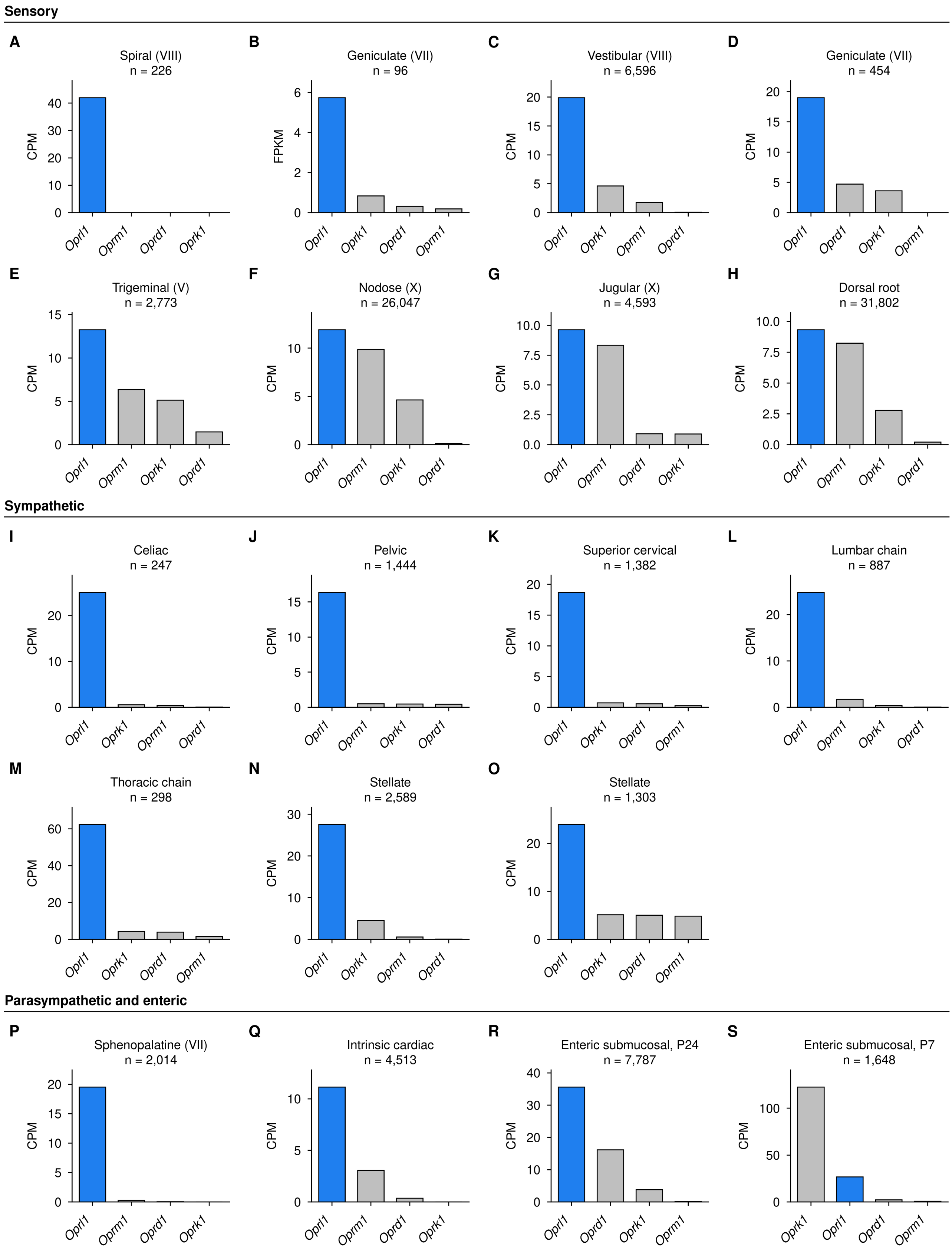


**Figure 1.** ***Oprl1*** **is the highest-expressed opioid receptor gene in peripheral neurons**

Mean expression of the four opioid receptor genes in each of 19 peripheral neuronal populations, grouped by division of the peripheral nervous system. Roman numerals give the cranial nerve of ganglia that belong to one. Bars are pseudobulk means over all neurons of the population, in the unit each dataset was measured in, ordered within each panel from highest to lowest. *Oprl1* is blue and the other three receptors are gray. Sample size is the number of neurons contributing to the panel.

(A) Spiral ganglion (VIII), GSE114997, full-length SMART-seq, median 2,678,701 reads per neuron. *Oprl1* 41.94 CPM in 74.8% of neurons; *Oprm1*, *Oprd1*, and *Oprk1* detected in no cell. All four genes are present in the annotation, so the three zeros are measured absences.

(B and D) Geniculate ganglion (VII) on two platforms from two laboratories. (B) GSE102443, full- length SMART-seq, FPKM, 6.92-fold over the second receptor (4.26 to 13.32), support 1.000, and 31.30-fold over *Oprm1*, which is the lowest of the four here. (D) GSE135801, 3' droplet, 4.02-fold, support 0.974 (95% interval 0.99 to 36.70).

(C) Vestibular ganglion (VIII), GSE309608, four mice, 4.29-fold (3.85 to 4.81), support 1.000, same ordering in each mouse.

(E) Trigeminal ganglion (V), iPain atlas, whole-cell neurons only, 2.08-fold (1.41 to 3.26).

(F and G) Inferior (nodose) and superior (jugular) ganglia of the vagus (X), NodoMap atlas, whole- cell datasets only. Nodose 1.21-fold (1.15 to 1.28), support 1.000, first in all four contributing datasets; jugular 1.16-fold (0.92 to 1.46), support 0.898, first in two of three.

(H) Dorsal root ganglion, a spinal ganglion with no cranial nerve, iPain atlas, whole-cell neurons only, 1.13-fold (1.08 to 1.19), the narrowest margin measured and the second largest sample.

(I to O) Sympathetic ganglia, ordered by margin. Celiac 44.41-fold (20.42 to 197.63), pelvic 32.42-fold (22.00 to 54.20), superior cervical 25.72-fold (15.93 to 52.14), lumbar chain 14.78-fold (12.00 to 18.83), thoracic chain 14.69-fold (9.35 to 27.27), stellate 6.08-fold (5.16 to 7.27), stellate 4.68-fold (3.02 to 8.08). Support 1.000 throughout. (I), (J), (L), and (N) are from GSE232789; (K) from the two untreated animals of GSE231766; (M) from GSE78845; (O) from GSE231924, which profiles cardiac-projecting neurons and places *Oprl1* first in each of eight mice. The stellate ganglion appears twice, in (N) and (O), from two deposits, two laboratories, and two neuron-calling routes. The pelvic ganglion carries both sympathetic and parasympathetic neurons and is grouped here with the sympathetic block.

(P and Q) Parasympathetic ganglia. (P) Sphenopalatine ganglion (VII), also called the pterygopalatine ganglion, GSE232789, 66.86-fold (40.54 to 132.54), the largest margin in the study. These neurons are cholinergic and not noradrenergic (*Slc18a3* 252 CPM, *Chat* 26, *Th* 2, *Dbh* 42) against *Th* 390 to 840 CPM in the sympathetic ganglia of the same experiment. (Q) Intrinsic cardiac nervous system, GSE330884, three mice, 3.65-fold (3.29 to 4.08), the only population in this division in which *Oprm1* is the second receptor. These ganglia are intrinsic to the heart and receive vagal preganglionic input, so they carry no cranial nerve numeral.

(R and S) Enteric submucosal neurons of the small intestine at two ages, GSE263422, one laboratory and one platform. (R) Postnatal day 24, *Oprl1* first at 2.20-fold (2.05 to 2.37) in all three samples. (S) Postnatal day 7, *Oprk1* first at 122.51 CPM against *Oprl1* 26.70, 4.59-fold (4.04 to 5.24) in both samples. *Oprk1* falls to 3.77 CPM at 1.1% detection by day 24 while *Oprl1* holds between 26 and 36 CPM.

All panels use whole-cell data. Both iPain atlases are majority single-nucleus and are restricted here to `suspension_type == "cell"`; pooling preparations compresses the trigeminal margin from 2.08-fold to 1.18-fold, for the reason given in Figure 2. Neurons were called on raw counts with glial and immune barcodes excluded, and every population passed the same marker gate and ambient- RNA check before any receptor value was read from it. Margins are the ratio of the highest to the second-highest receptor; intervals in parentheses are 95% bootstrap intervals over 10,000 resamples of the cells, and support is the fraction of those resamples retaining the observed top receptor. Panels (I), (J), (L), (N), and (P) come from one study of five autonomic ganglia in six samples, so laboratory and platform are constant across those five. That deposit was sequenced in two batches, stellate with sphenopalatine and the first pelvic sample, lumbar chain with celiac and the second, and median library ranges 2.6-fold across the five, from 27,931 to 73,326 UMI.

See also Figure 2, Figure S3, and Table S1.

## Nuclear libraries scale with gene length and whole-cell libraries do not

Single-nucleus libraries from vagal and dorsal root ganglia place *Oprm1* first, at 244.49 CPM against 5.22 for *Oprl1* and at 94.92 against 7.60, and the whole-cell fractions of those same atlases place *Oprl1* first (Figures 2A and 2B). The two preparations sample different RNA. A nucleus holds transcript that is still being made, unspliced and not yet exported, so most of a nuclear library is intronic; single-nucleus quantification counts reads across the entire gene body because counting exons alone would discard that majority. The number of reads assigned to a gene then scales with the length of its transcription unit as well as with the number of transcripts the neuron has made. Whole-cell libraries sample the cytoplasmic pool, which is the spliced mRNA available for translation, and 3' counting of that pool carries no length term. Across 14,876 protein-coding genes quantified in both preparations, nuclear expression rises with genomic span ($r = +0.403$) and whole-cell expression stays flat over a 68-fold range of length ($r = +0.022$; Figure 2E). *Oprm1* spans 279.7 kb and *Oprl1* spans 7.1 kb, giving nuclear-to-whole-cell ratios of 25.4 and 0.45 (Figure 2F), and *Oprm1* is detected in 71.9% of nuclei against 5.5% to 32.4% of cells (Figures 2C and 2D). The nuclear values report a combination of transcript abundance and gene length.

Several of the largest sensory ganglion atlases are single-nucleus, which accounts for the prevailing view that *Oprm1* is the dominant opioid receptor of peripheral neurons.

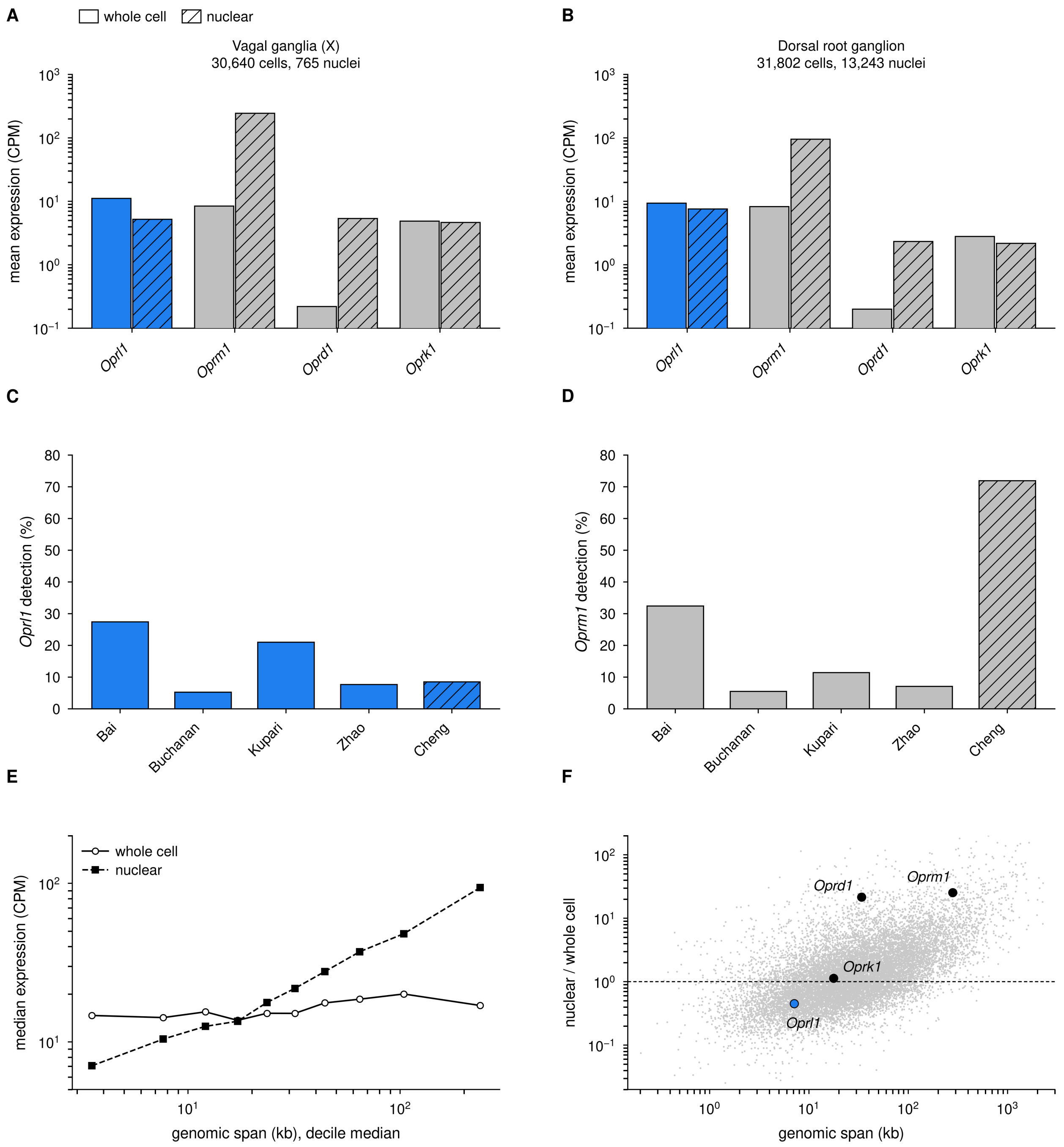


**Figure 2. A nuclear preparation reverses the receptor ordering**

Whole-cell and single-nucleus measurements of the same tissue, and the gene property that separates them. Solid bars are whole-cell data and hatched bars are nuclear data throughout, as the key above the panels gives. *Oprl1* is blue where the panel distinguishes genes by colour, as in Figure 1.

(A) Vagal ganglia (X), NodoMap. The whole-cell value is the unweighted mean over the atlas's four whole-cell deposits, 30,640 neurons; the nuclear value is the atlas's own 765 nuclei, the same tissue and the same integration. Whole cell: *Oprl1* 11.13 CPM, *Oprm1* 8.42, *Oprk1* 4.85, *Oprd1* 0.22. Nuclear: *Oprm1* 244.49, *Oprd1* 5.38, *Oprl1* 5.22, *Oprk1* 4.63. *Oprm1* rises 29-fold and *Oprl1* falls by half, which moves *Oprm1* from second to first.

(B) Dorsal root ganglion, a second tissue and a second laboratory. Whole cell, iPain atlas, 31,802 neurons: *Oprl1* 9.31 CPM, *Oprm1* 8.23, *Oprk1* 2.78, *Oprd1* 0.20. Nuclear, GSE201654 mouse arm, six samples and 13,243 nuclei: *Oprm1* 94.92, *Oprl1* 7.60, *Oprd1* 2.33, *Oprk1* 2.17. *Oprm1* is first in each of the six nuclear samples.

(C and D) Detection rate of *Oprl1* and of *Oprm1* in each of the five deposits that make up the NodoMap integration, on one scale. Deposits are named for the authors of the data. Bai, Buchanan, Kupari and Zhao are published cell suspensions that the NodoMap authors re-analysed; Cheng is the deposit the atlas labels in-house, the 765 nuclei the NodoMap authors generated themselves, and is the nuclear arm of (A) and (E) as well. (C) *Oprl1* is detected in 8.5% of the nuclei, inside the 5.3% to 27.5% range of the four whole-cell deposits. (D) *Oprm1* is detected in 71.9% of the same nuclei, against 5.5% to 32.4% in the whole-cell deposits, so it exceeds the highest whole-cell value by 2.2-fold. Detection rate scales with sequencing depth, which differs between deposits; (C) is what shows that depth does not account for (D), since the two genes were counted in the same libraries.

(E) Median expression against genomic span, over the 14,876 protein-coding genes of the atlas that reach 0.1 CPM in both preparations, binned into span deciles. One marker per decile, open circles for whole cell and filled squares for nuclear, joined to show the direction of each series. Genes were selected on expression and never on length, and the floor is low enough to admit all four opioid receptors. Whole-cell medians run between 13.7 and 20.0 CPM across deciles whose median spans run from 3.5 kb to 238.4 kb, a 68-fold range of length, and correlate with span at r = +0.022 over the individual genes. Nuclear medians rise from 7.1 to 94.1 CPM over the same deciles, r = +0.403. Gene length carries no information about how much mature transcript a neuron holds, so the preparation whose levels track it is the distorted one. The expression floor is a judgement call and does not carry the result: from no floor at all to 1 CPM, the whole-cell correlation runs from +0.117 to -0.065 and the nuclear one from +0.399 to +0.433 (`results/preparation_bias_floor_sensitivity.csv`).

(F) Nuclear level divided by whole-cell level, for each of those genes, against genomic span. Each grey point is one gene. The dashed line is no change, and the four opioid receptors are marked. *Oprl1* at 7.1 kb sits at 0.45, the 32nd percentile of the 2,431 genes within a factor of 1.6 of its length, and *Oprk1* at 17.7 kb sits at 1.14 and the 60th percentile of its own length class, so both behave as ordinary genes of their size and the ordering between them survives the preparation. *Oprm1* at 279.7 kb sits at 25.4 against a median of 4.94 for genes of comparable length, the 94th percentile of that class. *Oprd1* at 33.8 kb sits at 21.6 against a median of 1.24, the 99th percentile and the largest departure of the four; its whole-cell level is 0.25 CPM, so that ratio is measured on a small denominator and is the least certain of the four. For *Oprm1* and *Oprd1* length accounts for part of the gain and not all of it. Across the deciles of (E) the median ratio runs from 0.46 to 5.22 and crosses 1 at 15.8 kb. Both panels use the vagal neurons of (A), one atlas and one chemistry, so the contrast is between preparations rather than between studies. Spans are Ensembl GRCm39 gene loci.

A nucleus holds nascent transcript that has not been spliced or exported, and single-nucleus quantification counts reads across the whole gene body, introns included, since restricting to exons discards most of a nuclear library. Signal accrues with the length of the transcription unit rather than with the number of finished transcripts. Whole-cell libraries sample the cytoplasmic pool, the mature mRNA a neuron translates, and 3' counting of that pool has no length term. This is a bias in a known direction rather than a failure of the assay, and it reaches the opioid receptors because they span 7 to 280 kb.

Preparation is confounded with laboratory in (A) and (B), since no laboratory here has run both preparations on one tissue. (E) and (F) do not rest on that contrast: both are internal to a single atlas and a single chemistry, and they identify the biased measurement from the behaviour of 14,876 genes rather than from any receptor.

Every panel of Figure 1 therefore uses whole-cell data, and no human single-nucleus dataset can establish a species difference in the ordering: the mouse arm of the one cross-species nuclear experiment, shown in (B), reverses in the same direction.

## Discussion

Single-nucleus and whole-cell preparations of the same ganglia disagree about which opioid receptor predominates, and the disagreement follows gene length. The two preparations sample different RNA and capture it by different chemistry. A nucleus holds transcript that is still being made, unspliced and not yet exported, so most of a nuclear library is pre-mRNA; quantification counts reads across the entire gene body, because restricting to exons would discard that majority. Oligo(dT) primers hybridize to internal adenosine homopolymers as well as to the poly(A) tail. These homopolymers occur frequently in introns, and their number rises with the length of the transcription unit, so reads assigned to a gene scale with its genomic span as well as with the number of transcripts the neuron has made (Chamberlin et al., 2024). Mature mRNA carries a single priming site at its 3' end, and whole-cell libraries therefore record abundance without reference to length. Across 14,876 protein-coding genes, nuclear expression rises with genomic span (r = +0.403) while whole-cell expression stays flat over a 68-fold range (r = +0.022), and the four receptors order by span: *Oprm1* (279.7 kb) is enriched 25.4-fold in nuclei, *Oprd1* (33.8 kb) and *Oprk1* (17.7 kb) fall between, and *Oprl1* (7.1 kb) is depleted to 0.45. Span accounts for a minority of the variance in this ratio, and *Oprm1* lies above the trend for genes of comparable length. Gene length indexes pre-mRNA content imperfectly, and loci with long first introns, dual promoters and slow splicing, a description that fits *Oprm1* (Liu et al., 2021), accumulate nuclear signal beyond what span predicts; global length scaling will over-correct rapidly spliced genes and under-correct *Oprm1*. Dissociation offers a competing account, since it axotomizes every neuron and axotomy lowers opioid responsiveness while raising nociceptin responsiveness in DRG neurons (Abdulla and Smith, 1998), and that account predicts no transcriptome-wide length correlation. Ranking genes by CPM within a nuclear library confounds transcript abundance with transcription unit length. MOR primacy in peripheral analgesia rests on knockout phenotypes, reporter knock-in lines, radioligand binding and in situ hybridization (Scherrer et al., 2009; Wang et al., 2018); single-nucleus atlases have inflated that hierarchy rather than established it.

## Methods

### Data

No new sequencing was generated. All data are public single-cell or single-nucleus RNA-seq deposits, listed with their accessions, studies, platforms, sample sizes and quality-control outcomes in Table S1. Nineteen neuronal populations from 16 mouse ganglia and plexuses were drawn from 12 deposits published between 2016 and 2026, and were reached through GEO and CZ CELLxGENE.

Every value reported in Figure 1 comes from whole-cell libraries. Both iPain atlases and the NodoMap atlas contain nuclear as well as whole-cell fractions; those were restricted to `suspension_type == "cell"` before any receptor value was read, for the reason established in Figure 2. The nuclear fractions are used only in Figure 2, where they are the subject.

### Quantification

Expression is per-cell counts per million, computed as the count of a gene in a cell divided by that cell's total counts and multiplied by 10^6, and a population's level is the mean of that quantity over all its neurons. GSE102443 was deposited as FPKM and is analysed in that unit; levels are compared only within a population, never across units or platforms. GSE231924 was deposited as Seurat log-normalised values at a scale of 10,000, from which CPM is recovered as expm1(x) x 100, with the scale verified from the row sums rather than assumed.

A gene absent from a dataset's annotation is recorded as a null and never as a measured zero. Every level table carries an `in_matrix` field derived from the matrix itself, so that a reference gap and a measured absence remain distinguishable. The three zeros in the spiral ganglion are measured absences by that test: all four receptors are present in the annotation and three are detected in none of the 226 neurons.

### Cell filtering and neuron calling

Barcodes were retained at 2,000 or more UMI and 1,000 or more detected genes. Neurons were then called on raw counts, so that the threshold does not move with library size, and glial and immune barcodes were excluded on CPM, so that those thresholds do not move with sequencing run.

The neuronal criterion is *Snap25* at 5 counts or more together with a second marker chosen for the tissue: *Tubb3* for the autonomic and vestibular ganglia, *Th* for the superior cervical ganglion, *Phox2b* for the intrinsic cardiac nervous system, and *Elavl4* for the enteric plexus. Barcodes carrying a glial, myelinating or immune signature were excluded whatever their *Snap25* count, at *Sox10* below 200 CPM, *Plp1* below 500 CPM and *Ptprc* below 50 CPM. *Snap25* alone admits ambient-dominated barcodes, which is what these exclusions remove. Populations deposited as sorted or author-filtered neurons were taken as supplied.

### Marker gate

Before any receptor number was read from a population, its identity was confirmed against *Snap25*, *Phox2b*, *Slc17a6*, *Tac1*, *Calca* and *Actb*. *Snap25* and *Actb* are enforced: they must be present in the matrix and non-zero, and the pipeline raises rather than warns if they are not. The remaining markers are tissue-specific, and are recorded without being enforced. All 21 populations in the quality panel pass.

### Ambient RNA

No ambient correction is applied anywhere in this work, so a transcript read in neurons carries whatever the same transcript contributes to the surrounding soup. That contribution is measured instead of removed: within a single dissociation, each receptor's mean CPM in the called neurons is compared with its mean CPM in the non-neuronal barcodes captured beside them, as a log2 ratio with a pseudocount of 0.01 CPM. *Plp1* and *Ptprc* serve as negative controls.

*Oprl1*'s enrichment on its own conflates how neuronal the transcript is with how cleanly the two compartments separated in that dissociation. *Snap25* is neuronal by definition and calibrates the second, so the quantity reported is *Oprl1*'s enrichment minus *Snap25*'s. Across the eleven peripheral populations where both are available, *Oprl1* sits within 0.67 log2 of *Snap25*, median -0.14. The check ran on 14 of 21 populations; a deposit of sorted or author-filtered neurons has no non-neuronal compartment to score against, and the reason is recorded per population rather than left as a gap.

### Receptor ordering, bootstrap and sample agreement

The four receptors are ranked within a population, on the same cells and the same chemistry, which is what makes the ordering comparable between populations measured on different platforms. A rank is reported as determinate only when the top receptor is strictly above the runner-up. The margin is the ratio of the highest receptor to the second-highest.

Uncertainty in that ordering is estimated by resampling the cells of a population with replacement, 10,000 times, from a fixed seed. The 95% interval on the margin is the 2.5th and 97.5th percentile of the resampled margins, and support is the fraction of resamples in which the observed top receptor is still top. A pseudobulk ordering with a 1.8% margin and one with a 27% margin are otherwise reported identically, which is what this number prevents.

Where a deposit resolves into biological samples, each sample was also ordered on its own cells, and the count of samples placing the top receptor first is reported alongside the bootstrap.

### Whole-cell against nuclear preparation

Figure 2 compares the two preparations within the NodoMap integration: the same tissue, the same chemistry and the same annotation, so the contrast is between preparations rather than between studies. Preparation is nonetheless confounded with laboratory in the two paired tissue comparisons, since no laboratory here ran both preparations on one tissue, and the genome-wide analysis is what does not rest on that contrast.

Genomic spans are Ensembl GRCm39 gene loci, fetched once by symbol lookup so that every gene is measured on one annotation. Per-dataset feature tables carry coordinates for the features they quantify, but not for every gene and not on one assembly, which had previously dropped genes from the comparison and misplaced others.

Genes were kept if they are protein-coding by Ensembl biotype and reach 0.1 CPM in both preparations, which leaves 14,876. The floor exists only to keep ratios off a near-zero denominator; selection is on expression and never on length, and it is low enough to admit all four opioid receptors, which must not be filtered out of an analysis about them. Correlations are Pearson on log10 span against log10 expression, with Spearman on the untransformed values reported alongside. The floor was varied from none at all to 1 CPM: the whole-cell correlation runs from +0.117 to -0.065 and the nuclear one from +0.399 to +0.433, so the choice of floor sharpens the separation rather than creating it. Span deciles are ten equal-count bins of genes.

### Figures and tables

Figures are built to Cell Press artwork specification at the journal's fixed column widths, in Helvetica or a metric-compatible substitute, and are written at the size they will print rather than rescaled at submission. The build refuses to write a figure that exceeds the journal's width or height or that places any element outside the canvas. Table S1 is assembled from the analysis tables under `results/`; the build recomputes every margin from the levels Figure 1 is drawn from and refuses to write the table if a recorded margin disagrees with it.

### Software and availability

Analysis in Python 3.11 with pandas, numpy, scipy, anndata, h5py and matplotlib. All analysis code, the intermediate tables, and the scripts that produce every figure and table in this manuscript are in the repository accompanying this paper; the accession for every dataset is in Table S1.

## References


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## Supplemental information

### Table S1. The nineteen peripheral neuronal populations of Figure 1, related to Figure 1

One row per panel of Figure 1: the deposit each population was read from, the preparation and chemistry it was measured on, the neurons and biological samples contributing, the mean level of each of the four opioid receptor genes, and the statistics behind the ordering. Panel letters are those of Figure 1. The table the journal receives is `paper/tables/TableS1.csv`, one sheet of twenty-six columns; `paper/tables/TableS1.md` renders the same rows for reading, split into (a) provenance and sample and (b) levels, ordering and quality control. Both are written by `paper/tables.py` from the tables under `results/`.

Study and DOI name the publication that generated each deposit, taken from the deposit itself rather than from a search on the tissue: eleven of the twelve GEO records name their publication, either in the citation field or in the paper's own data availability statement. GSE309608 is the exception and is matched by its BioProject accession (PRJNA1335542), its six contributors and its title, all of which the PNAS paper carries. Full citations are in `paper/references.md`.

Divisions are the blocks of Figure 1. The pelvic ganglion carries both sympathetic and parasympathetic neurons and is grouped with the sympathetic block, as in the figure; `results/dataset_quality_panel.csv` records it as mixed autonomic. All nineteen populations are whole-cell, for the reason given in Figure 2; both iPain atlases are majority single-nucleus and are restricted here to `suspension_type == "cell"`.

Levels are pseudobulk means over all neurons of the population, in the unit each dataset was measured in: FPKM for the full-length geniculate deposit, CPM elsewhere. Levels are compared only within a panel, never across units or platforms. Detection is the percentage of that population's neurons in which *Oprl1* is called, and is reported per dataset for the same reason: it scales with sequencing depth, which differs by two orders of magnitude across these deposits. Median library is per neuron, in UMI for droplet deposits and reads for full-length ones.

Margins are the ratio of the highest receptor to the second-highest on the same cells. Intervals are 95% bootstrap intervals over 10,000 resamples of the cells, and support is the fraction of those resamples that retain the observed top receptor. Samples agreeing counts the biological samples that place the top receptor first on their own cells.

Neurons were called on raw counts with glial and immune barcodes excluded, and every population passed the same marker gate before any receptor value was read from it. The ambient-RNA check is the neuron-to-non-neuron ratio of each receptor within one dissociation; where it could not run, the table gives the reason rather than a blank, and those reasons are listed under the rendered table. A deposit of sorted or author-filtered neurons has no non-neuronal compartment to score against, which is what most of them are.

Blank cells are quantities the source tables do not record, and are left empty rather than carried over from a comparable population:

- **Samples** is recorded only for the populations `src/10_peripheral_ganglia.py` derived from source. For the rest, sample agreement is in the samples-agreeing column, which is why panel F reads 4/4 with no sample count beside it. `n/a` marks a deposit that does not resolve into biological samples here.
- **Median library** is not recorded for panels B, D, E, H and O. The two iPain populations were not re-derived from their source matrices, and the GSE231924 deposit carries no library sizes.
- **Detection** is not recorded for panels E and H, the two iPain populations, for the same reason.
- **Panel A** has no margin: the spiral ganglion quantifies all four receptors and detects one, so there is no second receptor to divide by. The three zeros are measured absences, not a reference gap.

## (a) Provenance and sample

| Panel | Division | Population | Deposit | Study | DOI | Platform | Preparation | Neurons | Samples | Median library |
|---|---|---|---|---|---|---|---|---|---|---|
| A | sensory | spiral (VIII) | GSE114997 | Shrestha et al., 2018 | 10.1016/j.cell.2018.07.007 | SMART-seq, full-length | whole cell | 226 | -- | 2,678,701 |
| B | sensory | geniculate (VII) | GSE102443 | Dvoryanchikov et al., 2017 | 10.1038/s41467-017-01095-1 | SMART-seq, full-length | whole cell | 96 | -- | -- |
| C | sensory | vestibular (VIII) | GSE309608 | Liu et al., 2026 | 10.1073/pnas.2530677123 | 10x droplet, 3' | whole cell | 6,596 | 4 | 55,172 |
| D | sensory | geniculate (VII) | GSE135801 | Zhang et al., 2019 | 10.1016/j.cell.2019.08.031 | droplet, 3' | whole cell | 454 | -- | -- |
| E | sensory | trigeminal (V) | iPain | Bhuiyan et al., 2024 | 10.1126/sciadv.adj9173 | 10x droplet, 3' | whole cell | 2,773 | -- | -- |
| F | sensory | nodose (X) | NodoMap | Cheng et al., 2026 | 10.1016/j.cpblue.2026.100072 | 10x droplet, 3' | whole cell | 26,047 | -- | 1,570 |
| G | sensory | jugular (X) | NodoMap | Cheng et al., 2026 | 10.1016/j.cpblue.2026.100072 | 10x droplet, 3' | whole cell | 4,593 | -- | 1,570 |
| H | sensory | dorsal root | iPain | Bhuiyan et al., 2024 | 10.1126/sciadv.adj9173 | 10x droplet, 3' | whole cell | 31,802 | -- | -- |
| I | sympathetic | celiac | GSE232789 | Sivori et al., 2024 | 10.7554/eLife.91576 | 10x droplet, 3' | whole cell | 247 | 1 | 56,023 |
| J | sympathetic | pelvic | GSE232789 | Sivori et al., 2024 | 10.7554/eLife.91576 | 10x droplet, 3' | whole cell | 1,444 | 2 | 40,986 |
| K | sympathetic | superior cervical | GSE231766 | Ziegler et al., 2023 | 10.1126/science.abn6366 | 10x droplet, 3' | whole cell | 1,382 | 2 | 9,885 |
| L | sympathetic | lumbar chain | GSE232789 | Sivori et al., 2024 | 10.7554/eLife.91576 | 10x droplet, 3' | whole cell | 887 | 1 | 73,326 |
| M | sympathetic | thoracic chain | GSE78845 | Furlan et al., 2016 | 10.1038/nn.4376 | full-length | whole cell | 298 | -- | 33,099 |
| N | sympathetic | stellate | GSE232789 | Sivori et al., 2024 | 10.7554/eLife.91576 | 10x droplet, 3' | whole cell | 2,589 | 1 | 27,931 |
| O | sympathetic | stellate | GSE231924 | Sharma et al., 2023 | 10.7554/eLife.86295 | 10x droplet, 3' | whole cell | 1,303 | -- | -- |
| P | parasympathetic and enteric | sphenopalatine (VII) | GSE232789 | Sivori et al., 2024 | 10.7554/eLife.91576 | 10x droplet, 3' | whole cell | 2,014 | 1 | 37,416 |
| Q | parasympathetic and enteric | intrinsic cardiac | GSE330884 | Xu et al., 2026 | 10.1016/j.cell.2026.06.040 | 10x droplet, 3' | whole cell | 4,513 | 3 | 52,533 |
| R | parasympathetic and enteric | enteric submucosal, P24 | GSE263422 | Li et al., 2025 | 10.1038/s41593-025-01962-x | 10x droplet, 3' | whole cell | 7,787 | 3 | 12,874 |
| S | parasympathetic and enteric | enteric submucosal, P7 | GSE263422 | Li et al., 2025 | 10.1038/s41593-025-01962-x | 10x droplet, 3' | whole cell | 1,648 | 2 | 8,974 |

## (b) Levels, ordering and quality control

| Panel | Population | Unit | Oprl1 | Oprm1 | Oprd1 | Oprk1 | Oprl1 detected (%) | Top | Runner-up | Margin | 95% CI | Support | Samples agreeing | Marker gate | Ambient check |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | spiral (VIII) | CPM | 41.94 | 0.00 | 0.00 | 0.00 | 74.8 | Oprl1 | none detected | undefined | -- | 1.000 | n/a | pass | not possible |
| B | geniculate (VII) | FPKM | 5.73 | 0.18 | 0.31 | 0.83 | 91.7 | Oprl1 | Oprk1 | 6.92x | 4.26 to 13.32 | 1.000 | n/a | pass | not possible |
| C | vestibular (VIII) | CPM | 19.88 | 1.78 | 0.09 | 4.63 | 57.6 | Oprl1 | Oprk1 | 4.29x | 3.85 to 4.81 | 1.000 | 4/4 | pass | run |
| D | geniculate (VII) | CPM | 18.98 | 0.01 | 4.72 | 3.61 | 19.6 | Oprl1 | Oprd1 | 4.02x | 0.99 to 36.70 | 0.974 | n/a | pass | not possible |
| E | trigeminal (V) | CPM | 13.24 | 6.36 | 1.46 | 5.12 | -- | Oprl1 | Oprm1 | 2.08x | 1.41 to 3.26 | 1.000 | n/a | pass | not possible |
| F | nodose (X) | CPM | 11.90 | 9.86 | 0.13 | 4.63 | 9.1 | Oprl1 | Oprm1 | 1.21x | 1.15 to 1.28 | 1.000 | 4/4 | pass | run |
| G | jugular (X) | CPM | 9.62 | 8.33 | 0.92 | 0.90 | 7.2 | Oprl1 | Oprm1 | 1.16x | 0.92 to 1.46 | 0.898 | 2/3 | pass | run |
| H | dorsal root | CPM | 9.31 | 8.23 | 0.20 | 2.78 | -- | Oprl1 | Oprm1 | 1.13x | 1.08 to 1.19 | 1.000 | n/a | pass | not possible |
| I | celiac | CPM | 25.03 | 0.42 | 0.03 | 0.56 | 66.4 | Oprl1 | Oprk1 | 44.41x | 20.42 to 197.63 | 1.000 | 1/1 | pass | run |
| J | pelvic | CPM | 16.33 | 0.50 | 0.43 | 0.46 | 51.0 | Oprl1 | Oprm1 | 32.42x | 22.00 to 54.20 | 1.000 | 2/2 | pass | run |
| K | superior cervical | CPM | 18.68 | 0.27 | 0.57 | 0.73 | 21.6 | Oprl1 | Oprk1 | 25.72x | 15.93 to 52.14 | 1.000 | 2/2 | pass | run |
| L | lumbar chain | CPM | 24.76 | 1.68 | 0.03 | 0.39 | 76.5 | Oprl1 | Oprm1 | 14.78x | 12.00 to 18.83 | 1.000 | 1/1 | pass | run |
| M | thoracic chain | CPM | 62.30 | 1.48 | 3.90 | 4.24 | 82.2 | Oprl1 | Oprk1 | 14.69x | 9.35 to 27.27 | 1.000 | n/a | pass | not possible |
| N | stellate | CPM | 27.55 | 0.56 | 0.03 | 4.53 | 52.5 | Oprl1 | Oprk1 | 6.08x | 5.16 to 7.27 | 1.000 | 1/1 | pass | run |
| O | stellate | CPM | 23.93 | 4.84 | 5.04 | 5.12 | 11.5 | Oprl1 | Oprk1 | 4.68x | 3.02 to 8.08 | 1.000 | 8/8 | pass | not possible |
| P | sphenopalatine (VII) | CPM | 19.50 | 0.29 | 0.04 | 0.01 | 49.9 | Oprl1 | Oprm1 | 66.86x | 40.54 to 132.54 | 1.000 | 1/1 | pass | run |
| Q | intrinsic cardiac | CPM | 11.13 | 3.04 | 0.36 | 0.00 | 38.2 | Oprl1 | Oprm1 | 3.65x | 3.29 to 4.08 | 1.000 | 3/3 | pass | run |
| R | enteric submucosal, P24 | CPM | 35.57 | 0.18 | 16.14 | 3.77 | 32.4 | Oprl1 | Oprd1 | 2.20x | 2.05 to 2.37 | 1.000 | 3/3 | pass | run |
| S | enteric submucosal, P7 | CPM | 26.70 | 0.74 | 2.29 | 122.51 | 23.4 | Oprk1 | Oprl1 | 4.59x | 4.04 to 5.24 | 1.000 | 2/2 | pass | run |

## Why an ambient check could not run

- (A) deposit is 226 sorted neurons, no non-neuronal cells
- (B) deposit is 96 sorted neurons, no non-neuronal cells
- (D) deposit is Phox2b-sorted neurons, no non-neuronal cells
- (E) source h5ad is 0.6 GB and was not retained; not re-derived
- (G) scored on the pooled ganglion, not the jugular subset
- (H) source h5ad is 20.9 GB, above this session's disk allowance
- (M) deposit is 298 sorted neurons, no non-neuronal cells
- (O) deposit is author-filtered neurons, no non-neuronal cells